How to justify your answer

- Formality: Casual conversation < Justification on an engineering exam <= Mathematical Proof
- Must be a logical argument that is

  Clear: Each step should follow immediately from previous ones

  Precise: Appropriate terms and symbols must be used correctly as defined in class, or a clear definition must be given

  Complete: All relevant cases and possibilities must be accounted for

  Correct: The justification must not contain any incorrect statements or logical steps

  Concise: The justification should not be needlessly long

- [Not graded, but useful in the real world: Insight and elegance – does the justification explain *why* the statement is true of false]

Justification Strategy

1) Write down relevant equations
2) Convince *yourself* whether the statement is true or false.
3) Write down the steps of your justification using words, equations, and/or numbers
   a) If the statement is true, try to prove directly
   b) If the statement is false, try to find a **counterexample** (or disprove directly)
4) Cross out any work that is not part of your justification